

PROJECT 1: CUSTOMER CHURN ANALYSIS FOR TELECOM INDUSTRY

PROJECT OVERVIEW

Objective: Predict churn and derive actionable strategies to retain users in a highly competitive telecom environment.

Duration: 2-week internship project

Status:  COMPLETE

KEY RESULTS

Overall Performance Metrics

Metric	Value
Total Customers Analyzed	5,000
Overall Churn Rate	34.24%
Churned Customers	1,712
Retained Customers	3,288
Model Accuracy (ROC-AUC)	75.82%

Critical Findings

 **Contract Type** is the #1 churn driver (14.8% importance) - Month-to-month: 51.6% churn rate - One-year: 25.5% churn rate - Two-year: 19.5% churn rate

 **New Customers (0-6 months)** show 56.6% churn rate - CRITICAL WINDOW

 **Complaints** directly correlate with churn: 2+ complaints = 65% churn

 **Satisfaction Threshold** at score 5: Below 5 = 70% churn, Above 8 = 5% churn

Risk Segmentation

Segment	Count	Churn Rate	Action
Loyal	307	11.7%	Retain & upsell
Low Risk	337	31.2%	Monitor
Medium Risk	316	54.7%	Targeted retention
High Risk	40	70.0%	Immediate intervention

DELIVERABLES

1. Data Files

- **telecom_customers.csv** - Complete dataset with 5,000 customers and 14 features
- **high_risk_customers.csv** - 40 high-risk customers requiring immediate attention
- **summary_statistics.csv** - Key metrics and statistics

2. Python Scripts

- **generate_telecom_data.py** - Data generation script with realistic churn drivers
- **churn_prediction_model.py** - ML model training and evaluation
- **create_visualizations.py** - Dashboard creation and visualization code

3. SQL Queries

- **churn_analysis.sql** - 11 SQL queries for data exploration and segmentation including:
 - Overall churn statistics
 - Churn by contract type, complaints, satisfaction
 - Tenure-based segmentation
 - High-risk customer identification
 - Loyalty analysis
 - Risk categorization

4. Machine Learning Models

- **logistic_regression_model.pkl** - Trained logistic regression model (ROC-AUC: 0.7582)
- **random_forest_model.pkl** - Trained random forest model (ROC-AUC: 0.7471)
- Both models can be loaded and used for prediction on new customers

5. Visualizations

- **churn_analysis_dashboard.png** - Comprehensive 9-panel dashboard showing:
 - Overall churn distribution (pie chart)
 - Churn by contract type, tenure, satisfaction
 - Churn by complaints, internet service, price segment
 - Service count impact, revenue vs churn analysis

6. Report & Documentation

- **CHURN_ANALYSIS_REPORT.md** - 2-page comprehensive report covering:
 - Introduction & Problem Statement
 - Executive Abstract with key findings
 - Tools Used (Python, SQL, ML libraries)
 - Complete methodology and steps
 - Key insights and actionable recommendations

- Business impact quantification
 - Conclusion with next steps
-

🔗 BUSINESS RECOMMENDATIONS

Priority 1: Contract Upgrade Program

- **Target:** Convert month-to-month customers to long-term contracts
- **Action:** Offer 25-30% discount for 2-year commitments
- **Expected Impact:** Reduce churn by 20-30% for converted customers
- **ROI:** Save \$277K annually from medium-risk segment

Priority 2: Early Engagement Strategy

- **Target:** Customers in first 6 months (56.6% churn rate)
- **Action:** Weekly check-ins, onboarding program, satisfaction survey at 3 months
- **Expected Impact:** Reduce early-stage churn to <40%

Priority 3: Complaint Resolution Excellence

- **Target:** Reduce complaints through SLA improvements
- **Action:** <4 hour response time, service credits for issues
- **Expected Impact:** -10% churn if complaints reduced by 50%

Priority 4: VIP Retention Program

- **Target:** 40 high-risk customers identified
 - **Action:** Dedicated account managers, monthly reviews, personalized offers
 - **Expected Impact:** Save 50% of high-risk segment = \$15.6K annual revenue
-

🔗 TECHNICAL STACK

Data Processing

- Python 3.x
- Pandas, NumPy
- SQL for data aggregation

Machine Learning

- Scikit-learn (Logistic Regression, Random Forest)
- Model evaluation metrics: ROC-AUC, Precision, Recall, F1-score
- Feature importance analysis

Visualization

- Matplotlib, Seaborn

- 9-panel dashboard with interactive insights
 - Ready for Power BI/Tableau integration
-

📊 MODEL PERFORMANCE

Logistic Regression (RECOMMENDED)

- **Accuracy:** 72%
- **Precision (Churned):** 64%
- **Recall (Churned):** 41%
- **ROC-AUC:** 0.7582 ★
- **Advantage:** Highly interpretable, clear feature coefficients

Random Forest

- **Accuracy:** 70%
 - **Precision (Churned):** 60%
 - **Recall (Churned):** 39%
 - **ROC-AUC:** 0.7471
 - **Advantage:** Better for non-linear patterns
-

🔍 WHAT'S INCLUDED FOR INTERVIEWS

Model Explainability

The analysis includes: - Feature importance rankings showing which factors drive churn - Logistic regression coefficients explaining positive/negative factors - Clear business interpretation of ML results

Customer Segmentation

- Risk-based segmentation (Loyal, Low-Risk, Medium-Risk, High-Risk)
- Actionable customer categories with specific interventions
- Churn probability for each customer

SQL Skills Demonstrated

- Complex queries with GROUP BY and aggregations
- CASE statements for segmentation
- Conditional filtering and ranking
- Multi-table analysis concepts

Python Skills Demonstrated

- End-to-end ML pipeline (data → model → evaluation)
 - Feature engineering and scaling
 - Model comparison and selection
 - Data visualization and insights
-

✦ TOP 50 INTERVIEW QUESTIONS COVERED

This project demonstrates proficiency with these key interview topics:

1. **Data Aggregation:** SQL GROUP BY, HAVING, window functions
 2. **Feature Engineering:** Creating derived features from raw data
 3. **Supervised Learning:** Classification models and evaluation metrics
 4. **Model Evaluation:** ROC-AUC, confusion matrix, precision-recall trade-offs
 5. **Feature Importance:** Understanding which factors drive predictions
 6. **Business Analytics:** Connecting models to business metrics and ROI
 7. **Customer Segmentation:** Creating actionable customer categories
 8. **Data Quality:** Handling imbalanced datasets (34% churn vs 66% retained)
 9. **Statistical Analysis:** Correlation analysis between variables
 10. **Hypothesis Testing:** Validating assumptions about churn drivers
-

🔗 NEXT STEPS

1. Deploy model in production for monthly scoring
 2. Set up automated alerts for high-risk customers
 3. Implement A/B testing on retention strategies
 4. Monitor model performance and retrain quarterly
 5. Track impact of retention programs on churn rate
 6. Scale successful tactics to other markets
-

🔗 HOW TO USE DELIVERABLES

For Quick Review

→ Start with: **CHURN_ANALYSIS_REPORT.md** (2 pages, 5-minute read)

For Technical Depth

→ Review: **churn_prediction_model.py** and **churn_analysis.sql**

For Visualizations

→ Check: **churn_analysis_dashboard.png** (9-panel dashboard)

For Interview Prep

→ Study: All files - they demonstrate end-to-end data science workflow

To Deploy Model

→ Load: **logistic_regression_model.pkl** or
random_forest_model.pkl

Project Completion Date: September 2026

Status: ✓ Ready for Production Deployment